



## Renan LAVIROTTE

MSc Computer Science  
Durham University

### Contact

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🐙 GitHub

🌐 LinkedIn

### Languages

French: Native  
English: Fluent  
Japanese: Intermediate

### Skills

#### Computer Science:

- Python
- C
- Unix Shell

#### Robotics & AI:

- Artificial Intelligence
- Machine Learning
- Computer Vision
- Sensor Fusion
- Embedded Systems
- Predictive Maintenance

#### Soft Skills:

- Teamwork
- Communication
- Project Coordination

### Interests

- Track & Field (national level)
- Tennis (15/3)
- Volleyball (regional level)
- Digital Design

### Profile

Multidisciplinary profile specialized in **autonomous robotics** and **artificial intelligence** with strong leadership and collaborative skills. As captain of a high-performance athletics team, I have led diverse groups in demanding environments, while my academic work demonstrates clear technical communication and team-based research. I am motivated to contribute rigorously, collaboratively, and responsibly to impactful, high-quality research.

### Education

#### Master's Degree at Durham University – UK

Oct 2024 – June 2028

- Key modules: AI/ML, Software Engineering, Paradigms, Theory of Computation
- Awarded First-Class Honours, 79% average in first year

### Experience & Achievements

#### Predictive Maintenance in C for Mars Rover

Nov 2025 – Dec 2025

- Extended memory safety across three levels (canaries, 40-byte alignment, and coalescing), ensuring mission-critical reliability of robotic systems.
- Developed an automatic post-radiation-storm recovery mechanism detecting 100% of bit-flip corruptions, strengthening the robustness of the robotic software stack and enabling reliable autonomous operations in harsh and unpredictable space environments.

#### Collaborative AI Bot – VR Environment (IBM Client)

Sept 2025 – Present

- Led the research and engineering of an AI bot in coordination with a five-person team (design, mechanics, AI) for a VR game, optimizing user experience and real-time adaptive interaction for maximum engagement.
- Integrated four IBM SkillsBuild AI modules delivering personalized feedback to support structured progression, improving user engagement, motivation, and overall satisfaction.

#### Environment-Reactive Autonomous Robot

July 2025 – Sept 2025

- Built an obstacle-avoidance autonomous robot (Python) through sensor integration, conducting 50+ systematic tests to optimize environmental perception.
- Combined hardware and software (servo motors, sensors) to enable continuous 180° scanning, allowing real-time trajectory adaptation and interaction with the environment.

#### Zero-Shot Multi-Modal Classification

Nov 2025 – Dec 2025

- Developed a zero-shot multimodal EEG paradigm using InfoNCE contrastive learning and CLIP semantic alignment, achieving 34.7% accuracy on five unseen classes and enabling cross-subject generalization without retraining.
- Conducted ablation and error analyses revealing interference between EEG and image modalities, motivating attention-based fusion for robust BCI applications with potential for assistive communication devices.

### Leadership & Responsibilities

#### Finalist – Artificial Heart Hackathon

Oct 2025 – Present

- Led sensor integration within an artificial heart prototype as part of a team of 40 engineers, evaluating trade-offs between accuracy, power consumption, and latency in critical environments.

#### Technical Collaborative Project Organization

Mar 2025 – Present

- Organized a hackathon bringing together 200+ participants, fostering ambitious technical projects and strengthening coordination and communication skills.

#### Captain – Athletics Club

Sept 2025 – Present

- Increased athlete engagement by over 20% within an elite team at a top-2 UK sports university competing in BUCS championships, promoting a culture of collective excellence.